Graduation Project Proposal

Project Information		
Project Title	**Department/Faculty/University**	**Project Field/Discipline**
V2X_Collision_Avoidance	Engineering – Electronics and Communication	Embedded System
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Is this project part of a mega-project at the same institution? ☐Yes ☐No (If yes, please submit all proposals together.)

If the project is sponsored by or initiated by an ICT company, please state its name:

Motivation

Please write why you chose this project idea, explaining clearly the problem that the project is addressing

Road safety and transportation efficiency are pressing global challenges, especially in rapidly developing countries like Egypt, where urban expansion and population growth have placed immense pressure on transport systems. Despite ongoing infrastructure improvements, road accidents still cause thousands of fatalities and injuries each year, along with significant economic losses.

Vehicle-to-Everything (V2X) communication offers a promising solution by enabling realtime interaction between vehicles and their surroundings: V2V (vehicle-to-vehicle), V2I (vehicle-to-infrastructure), V2P (vehicle-topedestrian), and V2N (vehicle-to-network). These connections support applications such as collision avoidance, adaptive traffic signals,

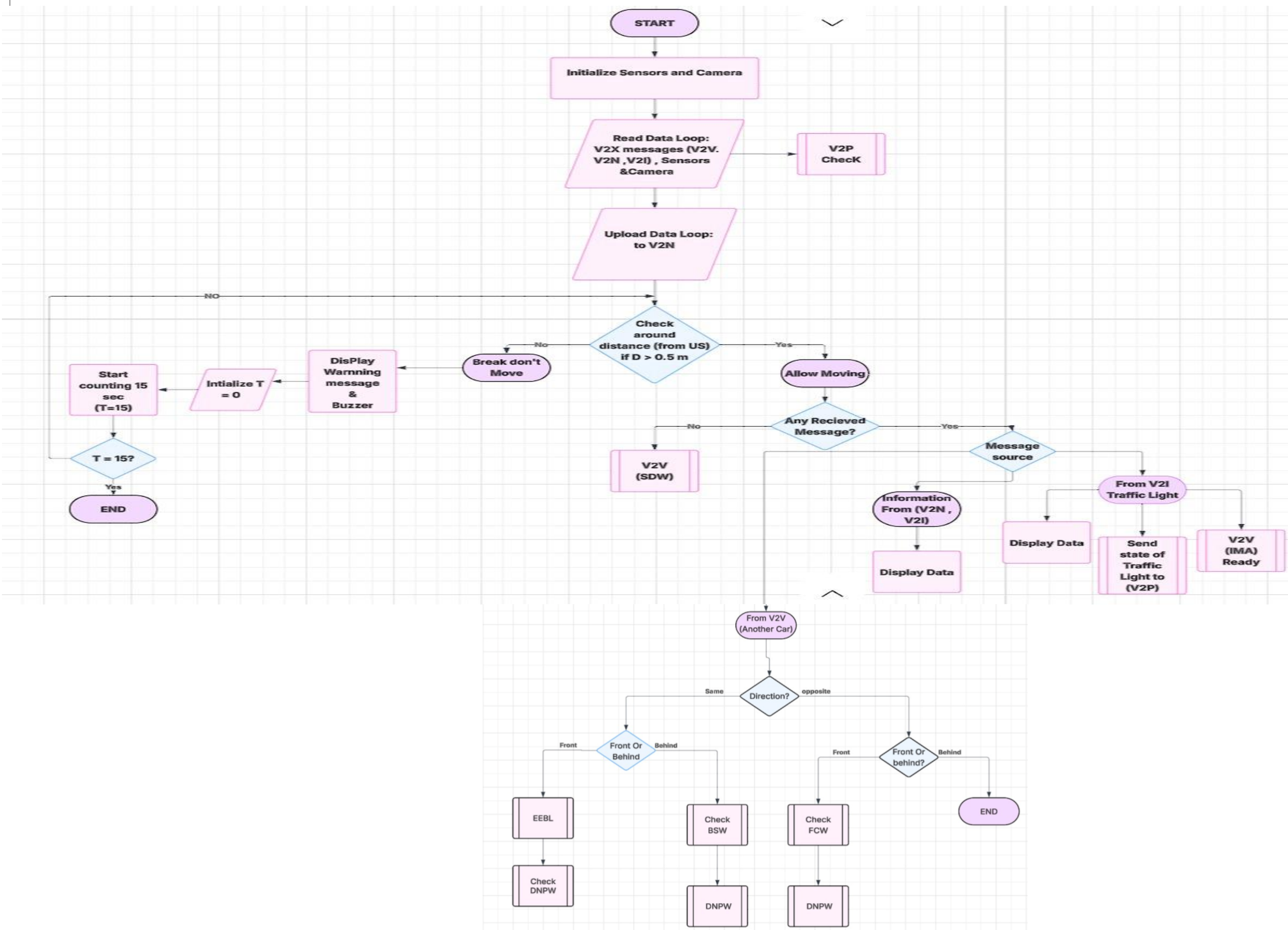
pedestrian safety alerts, and cooperative driving, with studies suggesting accident reductions of up to 80%.

Globally, V2X is gaining momentum for its role in enhancing safety, efficiency, and sustainability. For Egypt, adopting V2X can be transformative—saving lives, reducing economic losses, supporting smart city initiatives, and preparing the path toward autonomous and sustainable mobility.

Why do you think your project should be funded? for which the applicants write in a few lines where the help statement should be "Explain in no more than 3 lines the new and innovative aspects in your project that make it worthy of funding."

Our project stands out as an **integrated system that combines multiple intelligent safety subsystems into a single V2X solution**, rather than separate standalone projects. The system's reliability and performance depend heavily on **high-quality sensors and advanced microcontrollers** to ensure accurate detection and real-time response Consequently, the project involves **high material costs** to achieve the required level of quality and realistic implementation.

Block Diagram
Please insert the project detailed block diagram below, (Please highlight the parts that will be implemented in different colors than the parts that will be purchased)



Prototype Description and Specifications
Please note that ITAC only funds projects that result in a prototype. Include a clear description of how the prototype will operate, explaining a scenario/use case of the operation. Also include the performance metrics you target in the prototype.
The project proposes an intelligent AI-driven V2X collision avoidance system that continuously gathers real-time data from cameras and sensors, exchanges information with nearby vehicles and infrastructure, assesses collision risks, and instantly alerts the driver to prevent accidents, significantly enhancing road safety The performance of the proposed system will be evaluated based on the following key metrics: Detection Accuracy Measures the accuracy of the AI model in detecting vehicles, pedestrians, and obstacles under different traffic conditions. System Response Time Evaluates the time taken from data acquisition to issuing a warning, ensuring real-time operation. Collision Risk Detection Rate Assesses the system's ability to correctly identify potential collision scenarios. V2X Communication Reliability Measures the reliability and consistency of data exchange between vehicles and infrastructure. False Alarm Rate Evaluates the frequency of incorrect or unnecessary warnings issued by the system. System Stability and Continuity Assesses the system's ability to operate continuously without failure in real-world scenarios.
Project Plan
Please define the approach and phases to deliver the intended project outcome.

1-Requirements Analysis and Theoretical Study Phase *(First Semester)*

The project began with a comprehensive theoretical study of Intelligent Transportation Systems (ITS) and Vehicle-to-Everything (V2X) communication technologies During this phase, the following were carried out:

- Studying the concept of V2X and its various types (V2V, V2I, V2P, V2N).
- Analyzing traffic and accident problems in real-world scenarios, especially in developing countries like Egypt.
- Identifying user needs and system requirements based on these problems.
- Defining the project scope to be a prototype and simulation, rather than a full-scale real-world application.

This phase provided a strong scientific foundation for building the project.

- **2-System Design and Architecture Phase** ***(End of First Semester – Beginning of Second Semester)***

After completing the requirements analysis, **the project moved** to the system design phase, where the following were accomplished:

- Developing the overall design of the V2X system and dividing it into four main subsystems:
  Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I),Vehicle-to-Pedestrian (V2P),Vehicle-to-Network (V2N)
- Defining the function of each subsystem and the communication mechanism between them.
- Selecting appropriate hardware and software components.
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  This phase contributed to transforming theoretical ideas into a clear technical design.

3-Implementation and Development Phase ***(Second Semester)***

During this phase, the project systems **will be implemented** gradually, which will include:

- Developing vehicle safety systems such as:
- Collision warning
  Blind-spot monitoring
  Maintaining safe distance
-

  Implementing interaction scenarios with infrastructure, such as smart traffic signals.

- Applying the Vehicle-to-Network (V2N) system using a central server or cloud platform for real-time data exchange.
- Simulating data exchange between vehicles, infrastructure, and the network using appropriate communication protocols.

4-Testing and Performance Verification Phase

- After implementation, the system **will be tested** to ensure its efficiency and accuracy, through:
  Testing each subsystem individually.
- Verifying the correctness of data exchange and response times.
- Emergency situations

  Traffic congestion
  Collision risks
- Identifying potential issues and improving overall system performance.

This phase will help ensure the achievement of safety and efficiency objectives.

Prototype Prospects
List the Egyptian ICT companies that may be interested in the developed prototype and the end-users/customers (name the specific class of individuals, governmental agencies, ministries … etc. that will benefit from the prototype)

List of ICT Companies:

The developed V2X prototype integrates *Embedded Systems, Artificial Intelligence (AI), and IoT*, making it relevant to companies operating in intelligent transportation, smart city solutions, automotive technology, and traffic management. Potential Egyptian companies include:

- *Link Development* – specializes in smart city and IoT solutions.
- *Raya Smart Mobility / Raya Group* – involved in transportation and mobility solutions.
- *ITWorx* – provides AI and software integration for smart systems.
- *Orascom TMT / VINCI Solutions* – focuses on digital transformation and smart infrastructure.
- *Valeo Egypt* – automotive technology, including advanced driver assistance systems.

Potential End-Users/ Consumers:

The following *classes of individuals and organizations* in Egypt are expected to benefit from the V2X system:

- *Governmental Agencies / Ministries:*
- Ministry of Transport – for improving road safety and traffic management.
- Ministry of Interior – specifically traffic police units, for monitoring and reducing accidents.
- Ministry of Communications and Information Technology – for smart city integration and IoT infrastructure support.
- *Municipal and City Authorities:*
  Cairo Governorate and other urban municipalities – for traffic congestion management and infrastructure planning.
- *Private Sector and Fleet Operators:*
  Taxi, ridesharing, and logistics companies – to improve vehicle safety and operational efficiency.
- *Individual Road Users:*
  Drivers, pedestrians, and cyclists – who will receive real-time alerts and proactive safety measures through the V2V system

** Please fill in the budget form and upload both the proposal & budget forms to the student application form.